

Examining Relationship Status, Academic Performance, and Social and Dating Behavior at UCSD

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1 Introduction

The social environment at UC San Diego is often characterized by the stereotype that the campus is “socially dead,” a perception that circulates among prospective and current students and may even influence admissions by discouraging applicants who fear limited social or romantic opportunities. Despite how common this belief is, it has rarely been evaluated using quantitative evidence. This project aims to provide an empirical understanding of UCSD students’ social and dating behaviors by examining academic, demographic, and campus-involvement characteristics to see whether they meaningfully predict romantic involvement or relationship patterns. Through this analysis, we assess whether the “socially dead” stereotype reflects actual student experiences or whether it oversimplifies and misrepresents the diverse social realities on campus.

1.1 Research Questions

Our study is centered around two research questions:

RQ1: Can we predict the number of romantic involvements among UCSD students using academic and social factors?

This question focuses on understanding whether characteristics, such as major, year, GPA, campus involvement, and other attributes, serve as meaningful predictors of romantic activity. We approach this problem in two modeling stages by first predicting whether a student has ever been in a relationship, and then modeling the number of romantic involvements among those with at least one.

RQ2: For students who have been in a relationship, what factors are associated with the length of their longest relationship based on academic characteristics and social involvement?

This question investigates whether relationship duration can be explained by certain predictors. Because relationship length may reflect lifestyle stability, social availability, or time spent on campus, this question allows us to analyze how demographic and academic contexts relate to students’ relationship histories.

1.2 Data Collection

1.2.1 Survey Design

Data were collected using a Google Form designed to collect three categories of information:

Demographics (e.g., major, year, gender, race/ethnicity, transfer status, college affiliation, commute status, and optional GPA). These variables were included to identify potential confounders and to assess representativeness relative to the UCSD undergraduate population.

Campus involvement (e.g., participation in clubs, athletics, Greek life, campus employment). These served as independent variables, reflecting the assumption that higher engagement may correlate with larger social networks.

Relationship and social behavior, which served as the primary response variables. Students were asked about their relationship status, the number of romantic involvements they have had, the length of their longest relationship, long-distance status, whether they met partners at UCSD, and weekly social outings.

The survey was fully anonymous and voluntary, encouraging honest reporting while minimizing any hindrance to participation.

1.2.2 Sampling Method

We used convenience sampling, recruiting participants through, social media posts, Slack channels, in-person solicitation at Price Center, and sharing among friends and classmates. Although these methods were efficient for collecting responses quickly, this method does not produce a random or representative sample. Therefore, the generalizability of our findings is limited and must be interpreted with caution.

1.2.3 Biases

Several forms of bias were present in our sampling process:

Convenience bias: Students who often spend time in Price Center or follow project members on social media were more likely to participate, skewing the sample toward on-campus and more "chronically online" students.

Self-selection bias: Students who chose to participate may differ from those who did not. For example, those who are more open about their social lives or interested in topics related to relationships were more likely to participate.

Social desirability bias: Respondents may have underreported or overreported social activity or romantic involvement in ways they felt were socially acceptable.

Demographic imbalance: Certain majors, particularly Computer Science, Data Science, and Math-CS, were over-represented relative to the UCSD population. Third-years and on-campus residents were also disproportionately reflected.

2 Basic Analysis

2.1 Demographic Summary

A total of 110 UCSD undergraduates participated.

Overrepresentation of CS/DS/Math-CS majors (24.5% compared to 7% in the [UCSD Common Dataset 2024-2025](#)).

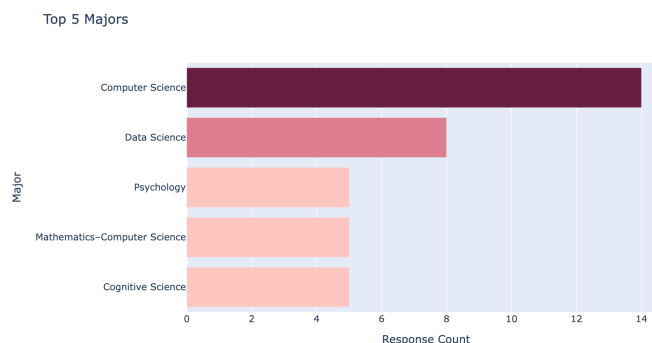


Figure 1: Major Distribution

Higher proportion of on-campus residents (38.2% off-campus vs. 52% off-campus in the [UCSD Common Dataset 2024-2025](#)).

Strong representation from third-year students.

These patterns reflect the convenience sampling approach and are important context when evaluating the generalizability of the models. At the same time, rather than viewing these gaps solely as limitations, they are seen as opportunities for future research. Expanding recruitment strategies, collaborating with additional student organizations, or conducting targeted outreach could help build a more representative dataset for a future research study.

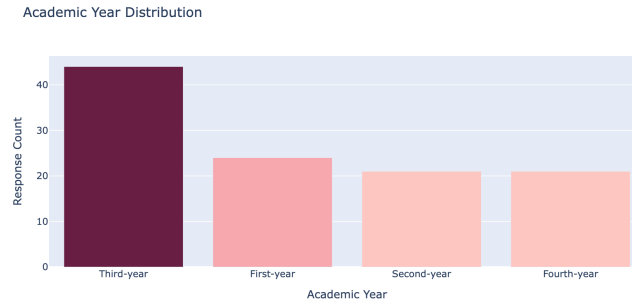


Figure 2: Academic Year Distribution

2.2 Missing Data Analysis

Some variables in the dataset contain missing values, which may impact analysis if not properly addressed. Table 1 summarizes the columns with missing data, sorted by the percentage of missingness.

Table 1: Columns with missing data.

Column	Missing Count	Missing %	Data Type
minor	74	67.3%	object
is_long_distance	49	44.5%	boolean
gpa	34	30.9%	float64
gender	3	2.7%	object

minor: Missing Not at Random, since not all students have a minor.

is_long_distance: Missing Not at Random, as not all students are in a relationship.

gpa: Missing Not at Random/Missing at Random, because the question was optional and some students may have chosen not to report. At the same time, a lack of a GPA may be correlated with year or transfer status.

gender: Missing Completely at Random, given the low proportion and no clear pattern.

2.3 Exploratory Data Analysis

Before addressing our two research questions, we conducted a series of exploratory analyses to understand the structure of our dataset and evaluate whether UC San Diego's academic and social characteristics meaningfully related to student's romantic histories.

The correlation matrix revealed that all continuous predictors were only weakly correlated with the target variables, suggesting early on that strong linear relationships were unlikely. Notably, the correlation for outings per week only exhibited a small positive association with the number of past romantic involvements but saw a negative association with the length of someone's longest relationship.

The distribution of the number of romantic relationships was highly right-skewed with a long tail containing students who had been in relationships significantly longer than the median respondent. This skew raised concerns about the suitability of linear modeling approaches and suggested the presence of potential subgroups within the population.

To establish a baseline, we used only the average of the training data. The naive baseline produced an MSE of 298 and an R^2 near zero, showing that even simple models struggled due to high variance in the target.

Overall, this preliminary exploration showed that continuous variables had low correlations with outcomes, relationship length exhibited strong skew and high variance, and baseline model performance was weak.

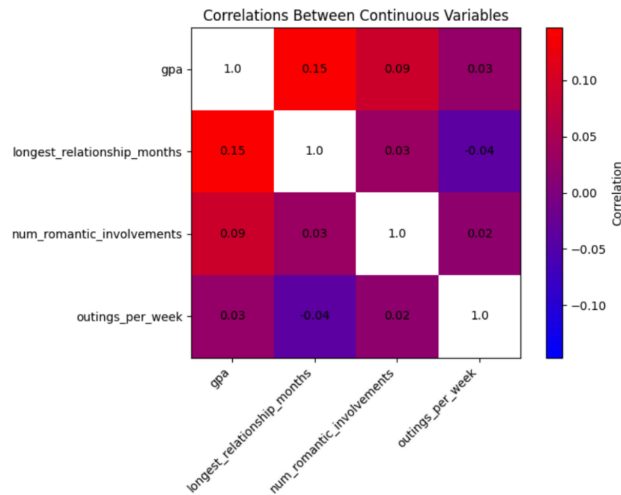


Figure 3: Correlation Matrix

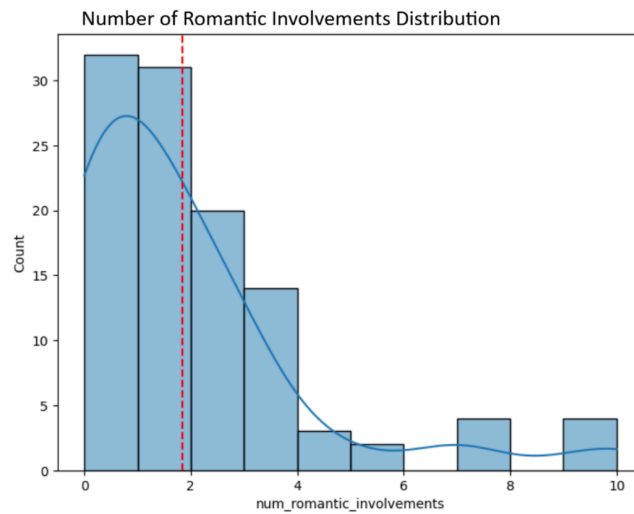


Figure 4: Distribution of Number of Romantic Involvements

3 Advanced Analysis

3.1 RQ1 Stage 1: Predicting Whether a Student Has Been in a Relationship

3.1.1 Method Overview

For the first stage of RQ1, we used logistic regression to predict whether a student has ever been in a relationship. The outcome variable was binary, with 0 representing having no relationships and >0 representing having at least one relationship. Logistic regression is well suited for this task because it is designed for binary outcomes and models the probability of an event occurring rather than predicting a continuous value.

We fit two separate models at this stage. The academic model focused on academic characteristics such as major and GPA, while the social model included social and behavioral factors. We evaluated each model using overall classification accuracy and examined coefficient estimates to understand which predictors were meaningful.

3.1.2 Academic Model Results

In the academic model, Arts & Humanities (A&H) major emerged as a statistically significant predictor of relationship status. The coefficient for A&H major was $\beta = -2.44$ with a p-value of .029, indicating a significant negative association.

In practical terms, this means that students majoring in Arts & Humanities were less likely to have been in a relationship compared to students in other majors, holding other academic factors constant. The negative coefficient reflects lower odds rather than a direct drop in probability, which is standard for logistic regression. Since the p-value is below .05, this result is unlikely to be due to random chance.

However, even with this significant predictor, the academic model only achieved an accuracy of 54%, which is just slightly better than guessing at random. This suggests that academic factors alone don't do a great job of predicting whether or not a student has been in a relationship.

3.1.3 Social Model Results

The social model did not produce any statistically significant predictors. None of the social variables included in the model showed a clear relationship with whether a student had been in a relationship.

Despite this, the social model achieved a higher overall accuracy of 60%. While this might initially seem like an improvement, the lack of significant coefficients suggests that the model is not capturing strong or reliable relationships between individual predictors and the outcome. Instead, the accuracy may be driven by broader patterns in the data rather than meaningful explanatory variables.

3.2 RQ1 Stage 2: Predicting Number of Romantic Involvements

3.2.1 Method Overview

Given the high proportion of zeros in our distribution of the target variable, `num_romantic_involvements`, we decided to filter our dataset to exclude values for this target which did not exceed zero as the next step in our analysis. To evaluate the predictive power of social and academic features for the number of romantic involvements of individuals who have had at least one relationship, we implemented a negative binomial regression model. This model was chosen because of its appropriateness for predicting count data with overdispersion, skew, and non-normality, characteristics which described our target variable exactly.

We extended the initial analysis paradigm from Part 1 and separated academic from social predictors to determine if the two categorical subtypes differed in their ability to predict our target. The predictor variables for the Social model were limited to club involvement, greek life membership, and number of outings per week. All predictors aside from `outings_per_week` were binarized.

The predictors for the Academic model were limited to minor, STEM classification, year, and major department. Again, all predictors for this model were binary, with year being one-hot encoded and `year_1` serving as the dummy variable. Major department was not one-hot encoded in order to accurately model double majors. Variables were vetted prior to model fit, where both values for each binary predictor were required to have > 5 observations.

3.2.2 Model Results

In consideration of the small N-size ($N = 78$) of our filtered dataset under the condition that $[(\text{number of relationships}) > 0]$, negative binomial regression models were fit on the full dataset to achieve the best point estimates using maximum likelihood estimators (MLE). This process essentially determines which coefficients maximize the probability of observing the data in our dataset. Only five predictors out of twenty-one showed statistical significance in their estimated coefficients: constant, `year_4`, and alpha for the Academic model, and constant and alpha for the Social model. The stability of each model and its coefficients were then assessed using the output of 500 bootstrapped resamples.

NegativeBinomial Regression Results						
Dep. Variable:	num_romantic_involvements	No. Observations:	78			
Model:	NegativeBinomial	Df Residuals:	66			
Method:	MLE	Df Model:	11			
Date:	Fri, 12 Dec 2025	Pseudo R-squ.:	0.04104			
Time:	16:23:46	Log-Likelihood:	-146.00			
converged:	True	LL-Null:	-153.18			
Covariance Type:	nonrobust	LLR p-value:	0.3221			
	coef	std err	z	P> z	[0.025	0.975]
const	1.5201	0.567	2.682	0.007	0.409	2.631
minor	-0.0148	0.193	-0.072	0.942	-0.393	0.365
is_stem	-0.5327	0.359	-1.484	0.138	-1.236	0.171
year_2	0.0761	0.282	0.270	0.787	-0.477	0.629
year_3	0.4081	0.241	1.695	0.090	-0.064	0.880
year_4	0.6200	0.302	2.055	0.040	0.029	1.211
dept_SocSci	-0.5341	0.533	-1.002	0.316	-1.579	0.510
dept_CogSci	-0.0918	0.628	-0.146	0.884	-1.322	1.138
dept_Engineering	-0.4207	0.668	-0.638	0.524	-1.714	0.873
dept_NaturalSci	-0.3007	0.603	-0.499	0.618	-1.402	0.800
dept_CompSci	-0.3722	0.600	-0.620	0.535	-1.548	0.804
dept_ArtsHum	-0.9276	0.484	-1.915	0.056	-1.877	0.022
alpha	-0.1515	0.075	2.019	0.044	0.004	0.299

NegativeBinomial Regression Results						
Dep. Variable:	num_romantic_involvements	No. Observations:	78			
Model:	NegativeBinomial	Df Residuals:	70			
Method:	MLE	Df Model:	7			
Date:	Fri, 12 Dec 2025	Pseudo R-squ.:	0.02302			
Time:	16:23:46	Log-Likelihood:	-149.66			
converged:	True	LL-Null:	-153.18			
Covariance Type:	nonrobust	LLR p-value:	0.4233			
	coef	std err	z	P> z	[0.025	0.975]
const	1.0380	0.237	4.377	0.000	0.573	1.503
club_professional	-0.1761	0.179	-0.983	0.326	-0.527	0.175
club_recreational	0.3168	0.207	1.534	0.125	-0.088	0.722
club_athletic	-0.1518	0.198	-0.766	0.444	-0.540	0.237
club_cultural	-0.2203	0.241	-0.913	0.361	-0.693	0.253
club_skilldev	0.1660	0.236	0.704	0.481	-0.296	0.628
is_greek	0.1693	0.253	0.669	0.504	-0.327	0.666
outings_per_week	-0.0200	0.045	-0.445	0.656	-0.108	0.068
alpha	0.1821	0.080	2.268	0.023	0.025	0.339

Figure 5: Academic Model Summary (left) Social Model Summary (right)

For both models, Academic and Social, all 500 bootstrapped draws issued convergence warnings despite the full models achieving convergence. This indicates model fragility, which could be a result of small N-size and sparse predictors. For instance, the size of some of the predictor variables was small (<10), meaning that they were potentially underrepresented in the bootstrapped samples. Additionally, given that only a handful of individuals were double-majors, if a resample of the Academic Model did not contain any double majors, the department columns would show collinearity.

Bootstrap finished: 500 successful fits out of 500				
Convergence warnings in 500 draws				
	MLE	Bootstrap mean	(MLE-Mean)/SD	Sign agreement (%)
const	1.520	1.434	0.198	99.8
minor	-0.014	-0.015	0.002	53.6
is_stem	-0.533	-0.507	0.067	90.6
year_2	0.076	0.069	0.023	59.0
year_3	0.408	0.402	0.024	92.6
year_4	0.620	0.604	0.050	96.4
dept_SocSci	-0.534	-0.495	0.096	89.8
dept_CogSci	-0.092	-0.064	0.056	54.0
dept_Engineering	-0.421	-0.496	0.118	77.2
dept_NaturalSci	-0.301	-0.276	0.051	73.0
dept_CompSci	-0.372	-0.353	0.045	81.6
dept_ArtsHum	-0.928	-0.892	0.110	99.6
alpha	0.152	0.082	1.060	100.0

Bootstrap finished: 500 successful fits out of 500				
Convergence warnings in 500 draws				
	MLE	Bootstrap mean	(MLE-Mean)/SD	Sign agreement (%)
const	1.038	0.995	0.150	99.8
club_professional	-0.176	-0.177	0.004	80.6
club_recreational	0.317	0.307	0.043	92.0
club_athletic	-0.152	-0.149	0.015	81.0
club_cultural	-0.220	-0.236	0.070	86.0
club_skilldev	0.166	0.156	0.039	74.4
is_greek	0.169	0.170	0.003	71.8
outings_per_week	-0.020	-0.011	0.167	63.6
alpha	0.182	0.131	0.858	100.0

Figure 6: Academic Model Bootstrap Results (left) Social Model Bootstrap Results (right)

In the assessment of coefficient stability, a couple red flags were raised in both models. The standard difference between the MLE coefficient values and the bootstrapped means was notably high for the dispersion parameter of both models, α , denoting a high sample bias for this estimate. Fortunately, all other coefficients showed moderate to high reliability between their bootstrapped mean and MLE. On the other hand, there seemed to be more uncertainty regarding the effect of each variable between samples. Sign agreement, a measure of consistency in the sign (+/-) of each coefficient estimate, was low ($<70\%$) for three coefficients in the academic model (minor, year_2, and dept_CogSci) and one in the social model (outings_per_week), suggesting sensitivity of these values to small changes in the data. Lastly, the coefficient distribution plots show a large 0 mass for the following predictors: minor, year_2, dept_CogSci, alpha (Academic Model), outings_per_week, and is_greek, indicating weakness of these predictors on the prediction of the target.

We examined each full model for its predictive power using residual plots. Both models seemed to be more likely to underestimate the number of relationships, but exhibited larger variance for overestimates. The Social model had more stable variance across estimates than the Academic model. The results of a 5-fold cross validation procedure generated an RMSE of 2.30 and 2.59 for the Academic and Social models respectively, elaborating on the results of the residuals plot. Given that our target variable has the most non-zero weight at low values, the RMSE values being greater than 2 highlights the minimally predictive nature of these models.

Finally, we visualized Observations vs. Predictions for each model. Predictions remained relatively constant across the range of observed values, with a variance of 0.60 for the Social model and 0.78 for the Academic model. In these plots, the failure of these models to make varying predictions despite changing features became immensely obvious. Though there was better performance of the Academic model given the RMSE, neither set of input variables was adept at predicting the number of romantic involvements of the participants.

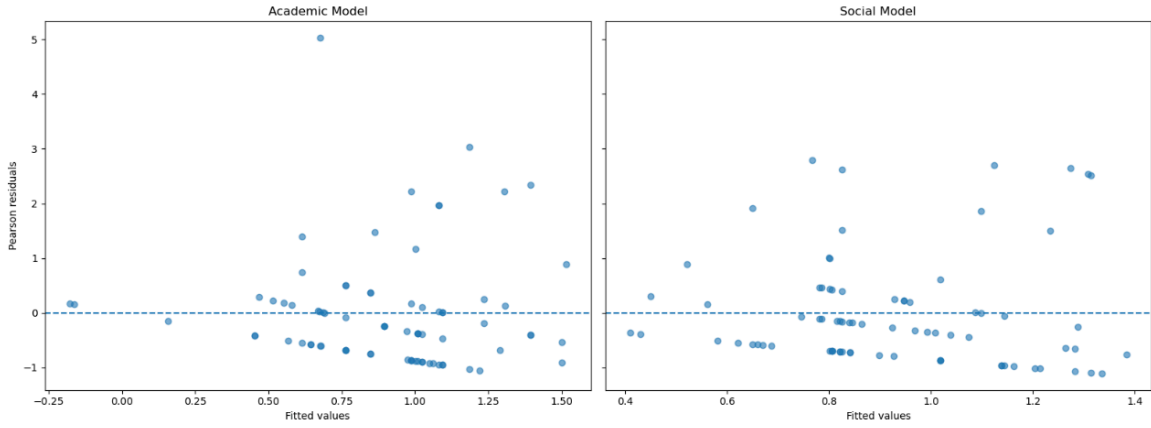


Figure 7: Academic Model Residual Plot (left) Social Model Residual Plot (right)

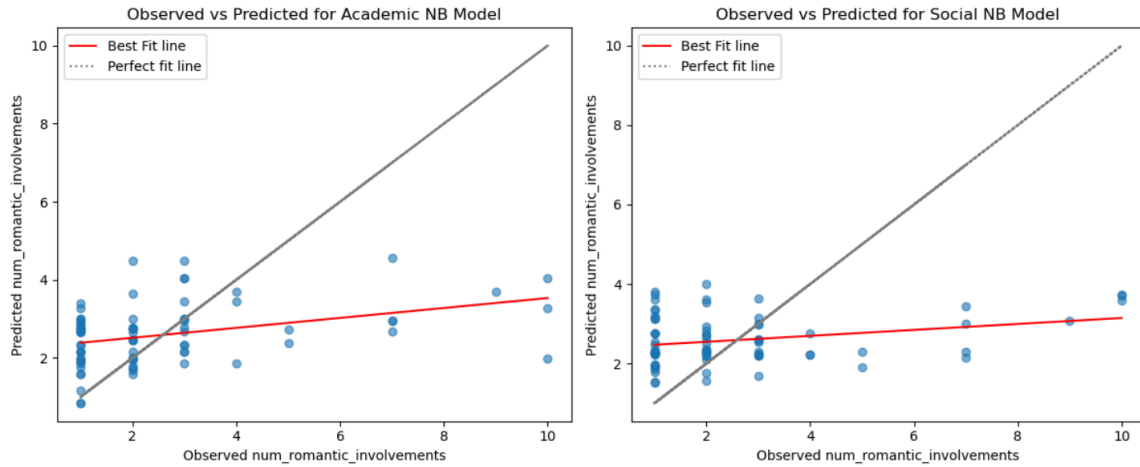


Figure 8: Academic Model Observed vs. Predicted (left) Social Model Observed vs. Predicted (right)

3.3 RQ2: Predicting Length of Longest Relationship

3.3.1 Method Overview

Having tried to predict if someone has been romantically involved before, we explored the factors that contribute to the length of an individual's longest relationship; past or present. We wanted to assess statistical significance of the 27 variables in our cleaned and preprocessed dataset. To do this, we first fit a linear model on our dataset, assessing significance by bootstrapping samples of 35 to create a distribution of mean squared errors from model predictions. This was compared to a naive prediction model, which always predicted the mean value of the target variable in our dataset. Concerned about overfitting, we used forward selection to choose 10 features from the original dataset to fit the linear regression model. The usefulness of these two models can be expressed in the following hypotheses:

H_0 : A linear regression model will not be better at predicting length of an individual's longest relationship than a naive mean prediction model.

H_A : A linear model with either every feature in our dataset or a subset of them will be a better predictor of length of longest relationship than the naive mean prediction model.

To test the hypothesis, we first approximated the sampling distribution of MSEs from the naive predictor and the linear models. Shuffling the dataset 10,000 times, a training and test set (20% the size of the full dataset) were chosen to fit every model before prediction.

3.3.2 Model Results

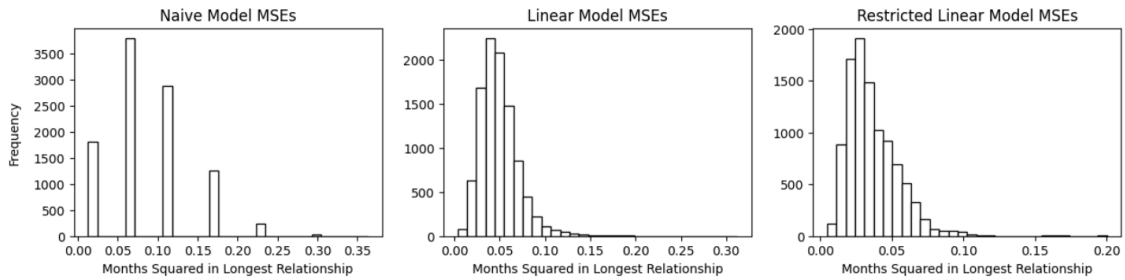


Figure 9: Distribution of MSEs

The naive predictor, full linear model, and selected linear model have mean MSEs of roughly 254.5, 418.8, 316.5 respectively. Conducting a t-test for difference in means, the linear model trained on all 27 features has a t-score of approximately -0.673, corresponding to a p-value of approximately 0.689. The linear model trained on only 10 features performs almost as poorly, with a t-score of about -0.241, which corresponds to a p-value of approximately 0.575. We therefore fail to reject the null hypothesis that a linear model is suitable for this data.

To take a more detailed look at the linear model, we fitted 10,000 linear models on bootstrapped samples of 35 to determine whether any variable coefficients were significant. We can see from the boxplots that 95% confidence intervals (represented by the whiskers) of every variable coefficient of all contain zero, indicating that none of the variables we collected are statistically significant predictors of the length of someone's longest relationship. Statistical significance remains unchanged when a subset of dataset features are used for modeling. This means that the linear model fitted on the data we have is not a suitable model for predicting how many months an individual's longest relationship has been.

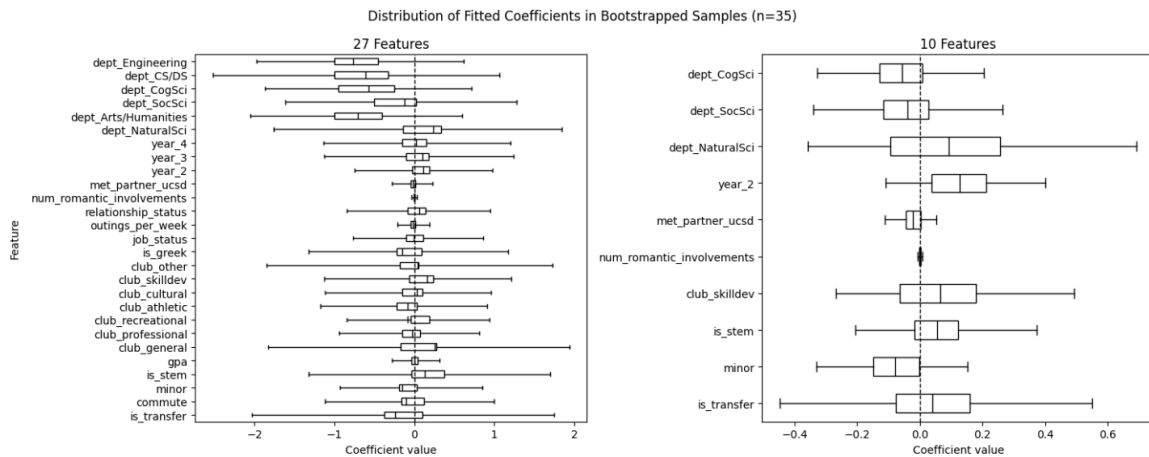


Figure 10: Distribution of Fitted Coefficients in Bootstrapped Samples

With this in mind, we tried exploring if we could detect latent variables, such as attitude towards relationships, by using unsupervised clustering based on academic and social factors. Both predicting cluster means for an individual and modeling every cluster separately yielded no significant results, and given the size of our study, overfitting is likely. We conducted a permutation test using a linear regression model with groups assigned according to cluster to determine whether statistically significant differences in means emerged from unsupervised groupings of individuals. The results showed a p-value of about 0.703, whose sampling distribution may change depending on the clustering algorithm used.

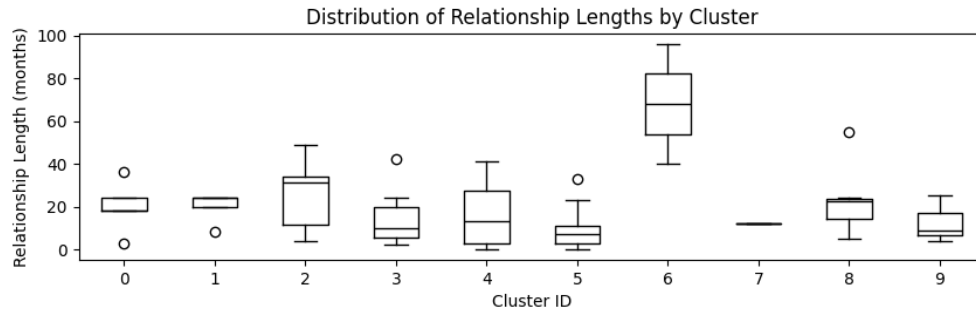


Figure 11: Distribution of Relationship Lengths by Cluster

4 Conclusion

4.1 Implications of Results

4.1.1 RQ1: Predicting Past Relationship Count

Using logistic regression, we found that the Arts/Humanities department was the only statistically significant predictor ($p = 0.029$). All other academic and social variables were not statistically significant enough ($p > 0.05$) in predicting past relationship counts. In other words, the results from Stage 1 suggest that major, particularly Arts & Humanities, may have some association with relationship experience, but the effect is relatively weak. Social variables, at least as modeled here, do not provide clear predictive power. While logistic regression allows us to interpret coefficients directly, the models' limited accuracy indicates that relationship experience is likely influenced by more complex factors than those captured in this stage. These findings suggest the need for more flexible models and richer representations of relationship history in later stages of analysis.

However, when comparing sets of predictors, social characteristics (e.g., club involvement, Greek life) showed roughly a 6% improvement in predictive ability compared to academic variables alone (60% for social predictors to 54% for academic predictors). This likely occurs because social factors exhibit lower variance and align more closely with interpersonal behavior than academic metrics such as GPA or major.

4.1.2 RQ2: Predicting Relationship Length

The naive model's bimodal MSE distribution revealed that relationship length likely arises from two distinct subpopulations with different expected durations. We applied linear models, naive estimators, and clustering to explicitly model that subgroup. Although clustering produced a slightly positive R^2 (0.0097), which was an improvement over the negative R^2 values of simpler models, the gains were minimal and became unstable as the cluster count increased. Relationship length in our data contained a strong skew, high variance, and multimodal structure that prevented any global model from making stable or reliable predictions. While social characteristics were marginally better predictors than academic ones, neither set of variables was effective. Predicting romantic history using broad academic and social indicators appears far more nuanced than our current dataset can capture.

4.2 Limitations

Our analysis revealed that both academic and social factors were poor predictors of past relationship counts and relationship length, but several limitations likely contributed to these weak results. A significant limitation was due to the sample size being extremely small ($n = 110$), and the dataset suffered from convenience sampling, social desirability bias, and demographic imbalances (specifically an overrepresentation of Computer Science and Data science students relative to other majors at UC San Diego). This uneven distribution limited the generalizability of our findings and may have amplified variance within key predictors. Additionally, collinearity among academic and social variables likely weakened model interpretability, and the right-skewed, high-variance outcome distributions reduced the effectiveness of linear models, causing instability in predictions across multiple modeling approaches.

4.3 Future Work

Future Work would start with expanding the dataset through continued data collection to increase the sample size while using stratified random sampling across department, college, or year to address the demographic imbalances present in our current data. Collecting richer forms of data, such as continuous behavioral measures, time-series information, or more nuanced social and academic engagement indicators may capture elements of romantic behavior that simple academic and social variables cannot. Applying more flexible modeling techniques, including tree-based models, zero-inflated models, or neural networks could help identify nonlinear or latent patterns missed by our current approaches. Together, these improvements would allow for a deeper and more accurate understanding of how academic and social factors shape the romantic lives of UC San Diego students.

4.4 Contributions

All group members collaborated on brainstorming the topic, designing the survey, and recruiting participants. Snigdha handled data cleaning and preprocessing. Sophie conducted the statistical analysis for Research Question 1. Arya led the modeling for Research Question 2. Susana and Karina supported analysis, created figures, and contributed to writing. All members helped with exploratory data analysis, writing the final report, and preparing the presentation. Generative AI was used for debugging and general formatting assistance.